Ehrenfeucht-Fraisse equivalence: n-pe games capture quantifier-rank n formulas

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-model-theory-basics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the parenthetical convention that a tuple satisfies A via any matching assignment is inside the statement span; 'Similarly one shows' closes the symmetric game clause verbatim.

Editorial note. The label thm:b-n-f names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Finiteness of quantifier-rank n types over a finite relational language. Same part, \ollabel{prop:qr-finite} (source line 1062; the author's own proof of that proposition is a bare 'by induction on n'). Line 1123: the converse direction's bookkeeping that only finitely many inequivalent formulas exist at each rank, letting the type T^a_n be condensed into a single formula.
- Definition of the n-pe game relation I_n. Same section: the pebble-game equivalence definition preceding the theorem. Everything the induction proves is about this relation; the game rules themselves are imported by reference.

Source reference. Original proposition prop-qr-finite: sources/olp-model-theory-basics/pinned-original-source.tex, source lines 1062–1062. Prerequisite pointer invoked at proof line(s) 1121; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *Let $\mathcal{L}L$ be a purely relational language. Then $I_n(\mathbf{a}, \mathbf{b})$ implies that for every A such that $\operatorname{qr} A \leq n$, we have $\mathfrak{M}, \mathbf{a} \vDash A$ if and only if $\mathfrak{N}, \mathbf{b} \vDash A$ (where again $\mathbf{a}$ satisfies A if any s such that $s(x_i) = a_i$ satisfies A). Moreover, if $\mathcal{L}L$ is finite, the converse also holds.*

Proof. The proof that $I_n(\mathbf{a}, \mathbf{b})$ implies that $\mathbf{a}$ and $\mathbf{b}$ satisfy the same formulas of quantifier rank no greater than n is by an easy induction on A. For the converse we proceed by induction on n, using prop-qr-finite, which ensures that for each n there are at most finitely many non-equivalent formulas of that quantifier rank.

For $n = 0$ the hypothesis that $\mathbf{a}$ and $\mathbf{b}$ satisfy the same quantifier-free formulas gives that they satisfy the same atomic ones, so that $I_0(\mathbf{a}, \mathbf{b})$.

For the $n + 1$ case, suppose that $\mathbf{a}$ and $\mathbf{b}$ satisfy the same formulas of quantifier rank no greater than $n + 1$; in order to show that $I_{n+1}(\mathbf{a}, \mathbf{b})$ suffices to show that for each $a \in |\mathfrak{A}|M$ there is a $b \in |\mathfrak{A}|N$ such that $I_n(\mathbf{a}a, \mathbf{b}b)$, and by the inductive hypothesis again suffices to show that for each $a \in |\mathfrak{A}|M$ there is a $b \in |\mathfrak{A}|N$ such that $\mathbf{a}a$ and $\mathbf{b}b$ satisfy the same formulas of quantifier rank no greater than n.

Given $a \in |\mathfrak{A}|M$, let T_n^a be set of formulas $B(x, \mathbf{y})$ of rank no greater than n satisfied by $\mathbf{a}a$ in $\mathfrak{M}$; T_n^a is finite, so we can assume it is a single first-order formula. It follows that $\mathbf{a}$ satisfies $\exists x\, T_n^a(x, \mathbf{y})$, which has quantifier rank no greater than $n + 1$. By hypothesis $\mathbf{b}$ satisfies the same formula in $\mathfrak{N}$, so that there is a $b \in |\mathfrak{A}|N$ such that $\mathbf{b}b$ satisfies T_n^a; in particular, $\mathbf{b}b$ satisfies the same formulas of quantifier rank no greater than n as $\mathbf{a}a$. Similarly one shows that for every $b \in |\mathfrak{A}|N$ there is $a \in |\mathfrak{A}|M$ such that $\mathbf{a}a$ and $\mathbf{b}b$ satisfy the same formulas of quantifier rank no greater than n, which completes the proof. □